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# **basic education**

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Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

## **SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS**

**MATHEMATICS P2**

**2022**

**MARKS: 150**

**TIME: 3 hours**

**This question paper consists of 14 pages and 1 information sheet.**

**INSTRUCTIONS AND INFORMATION**

Read the following instructions carefully before answering the questions.

1. This question paper consists of 10 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers correct to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. An information sheet with formulae is included at the end of the question paper.
9. Write neatly and legibly.

**QUESTION 1**

The table below shows the mass (in kg) of the school bags of 80 learners.

| MASS (in kg)     | FREQUENCY |
|------------------|-----------|
| $5 < m \leq 7$   | 6         |
| $7 < m \leq 9$   | 18        |
| $9 < m \leq 11$  | 21        |
| $11 < m \leq 13$ | 19        |
| $13 < m \leq 15$ | 11        |
| $15 < m \leq 17$ | 4         |
| $17 < m \leq 19$ | 1         |

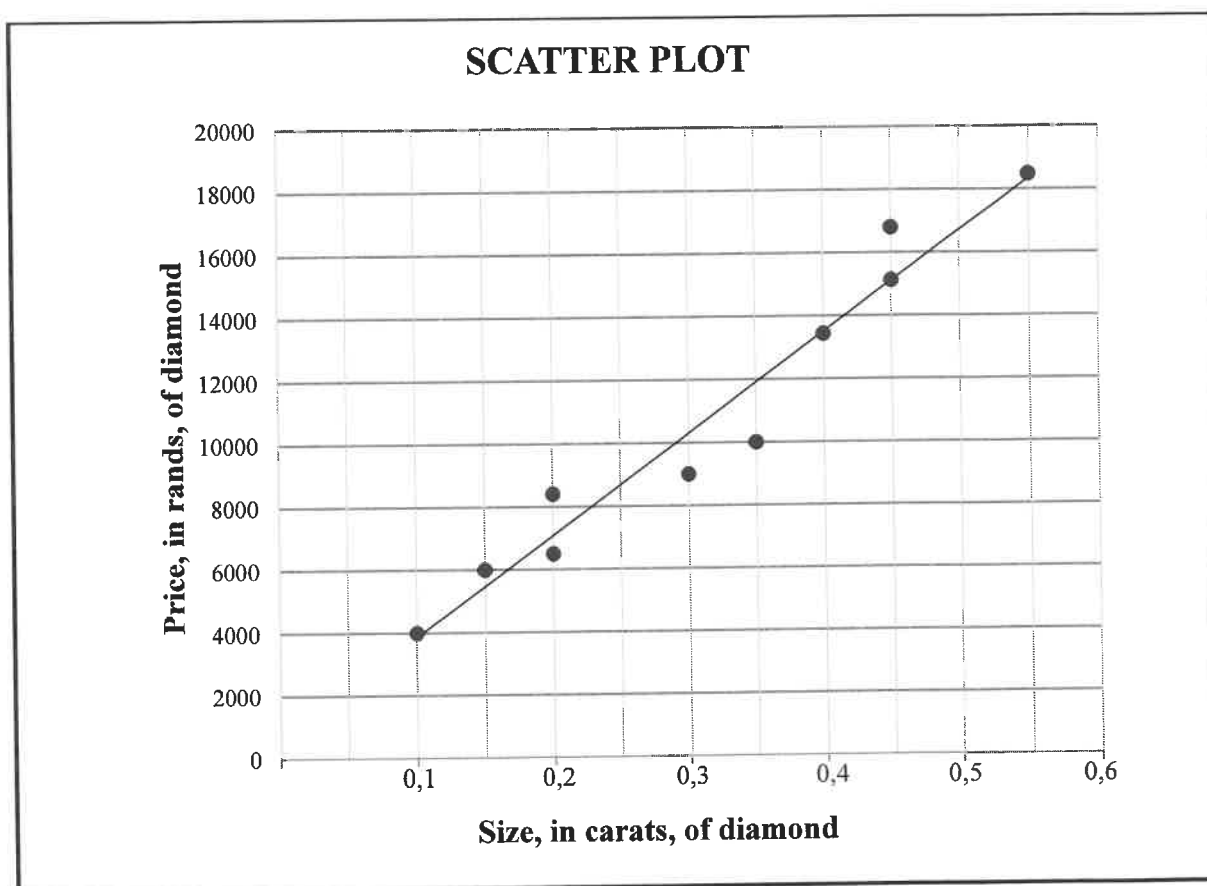
- 1.1 Write down the modal class of the data. (1)
- 1.2 Complete the cumulative frequency column in the table in the ANSWER BOOK. (2)
- 1.3 Draw a cumulative frequency graph (ogive) for the given data on the grid provided in the ANSWER BOOK. (3)
- 1.4 Use the graph to determine the median mass for this data. (2)
- 1.5 The international guideline for the mass of a school bag is that it should not exceed 10% of a learner's body mass.
  - 1.5.1 Calculate the estimated mean mass of the school bags. (2)
  - 1.5.2 The mean mass of this group of learners was found to be 80 kg. On average, are these school bags satisfying the international guideline with regard to mass? Motivate your answer. (2)

**[12]**

**QUESTION 2**

The table below shows the size (in carats) and the price (in rands) of 10 diamonds that were sold by a diamond trader. This information is also presented in the scatter plot below. The least squares regression line for the data is drawn.

|                                        |       |       |       |       |       |        |        |        |        |        |
|----------------------------------------|-------|-------|-------|-------|-------|--------|--------|--------|--------|--------|
| <b>Size, in carats, of diamond (x)</b> | 0,1   | 0,15  | 0,2   | 0,2   | 0,3   | 0,35   | 0,4    | 0,45   | 0,45   | 0,55   |
| <b>Price, in rands, of diamond (y)</b> | 4 000 | 6 000 | 6 500 | 8 400 | 9 000 | 10 000 | 13 440 | 15 120 | 16 800 | 18 480 |

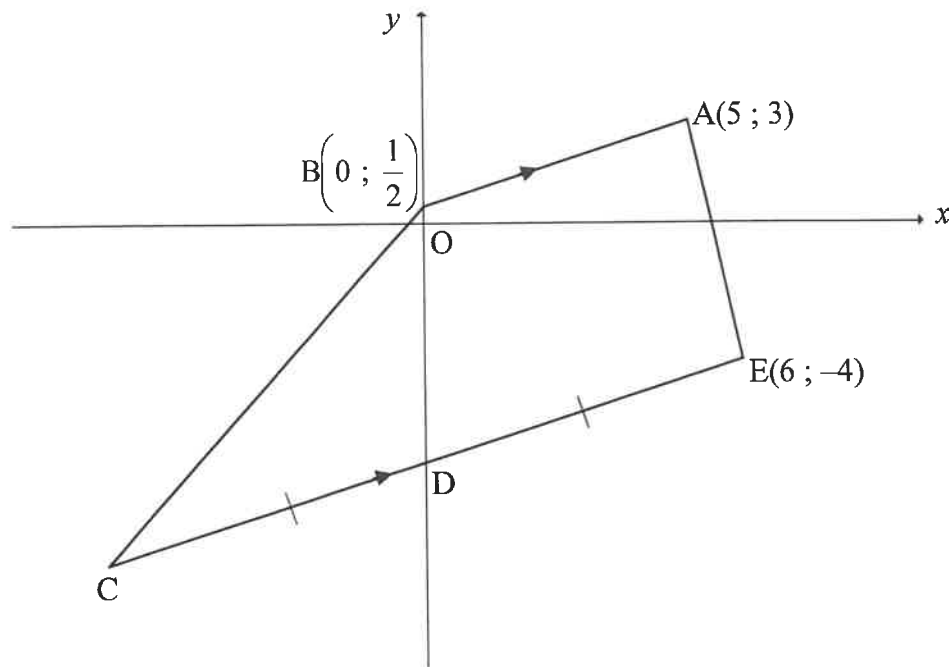


- 2.1 Determine the equation of the least squares regression line for the data. (3)
- 2.2 If the trader sold a diamond that was 0,25 carats in size, predict the selling price of this diamond in rands. (2)
- 2.3 Calculate the average price increase per 0,05 carat of the diamonds. (2)
- 2.4 It was later found that the selling price of the 0,35 carat diamond was recorded incorrectly. The correct price is R11 500. When this correction is made to the data set, the correlation between the size and price of these diamonds gets stronger. Explain the reason for this by referring to the given scatter plot. (1)

**[8]**

**QUESTION 3**

In the diagram,  $A(5; 3)$ ,  $B\left(0; \frac{1}{2}\right)$ ,  $C$  and  $E(6; -4)$  are the vertices of a trapezium having  $BA \parallel CE$ .  $D$  is the  $y$ -intercept of  $CE$  and  $CD = DE$ .

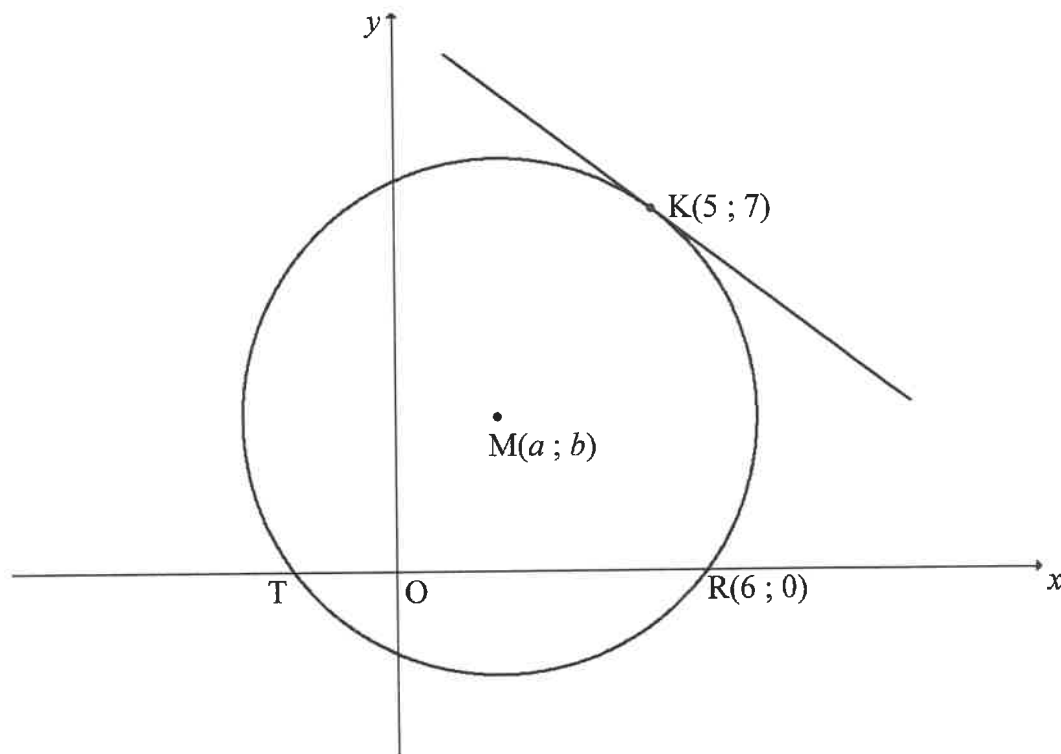


- 3.1 Calculate the gradient of  $AB$ . (2)
- 3.2 Determine the equation of  $CE$  in the form  $y = mx + c$ . (3)
- 3.3 Calculate the:
- 3.3.1 Coordinates of  $C$  (3)
- 3.3.2 Area of quadrilateral  $ABCD$  (4)
- 3.4 If point  $K$  is the reflection of  $E$  in the  $y$ -axis:
- 3.4.1 Write down the coordinates of  $K$  (2)
- 3.4.2 Calculate the:
- (a) Perimeter of  $\triangle KEC$  (4)
- (b) Size of  $\hat{KCE}$  (3)

**[21]**

**QUESTION 4**

In the diagram, the circle centred at  $M(a; b)$  is drawn.  $T$  and  $R(6; 0)$  are the  $x$ -intercepts of the circle. A tangent is drawn to the circle at  $K(5; 7)$ .

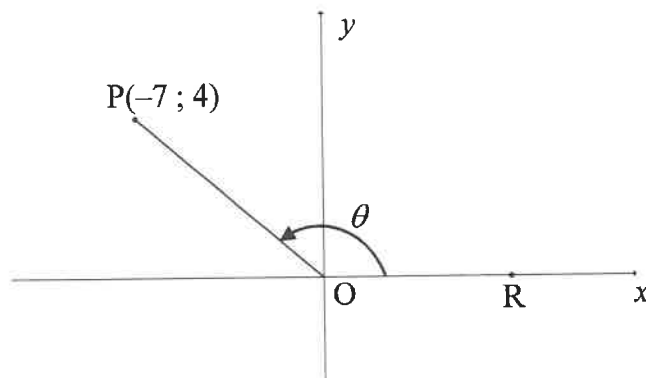


- 4.1  $M$  is a point on the line  $y = x + 1$ .
- 4.1.1 Write  $b$  in terms of  $a$ . (1)
- 4.1.2 Calculate the coordinates of  $M$ . (5)
- 4.2 If the coordinates of  $M$  are  $(2; 3)$ , calculate the length of:
- 4.2.1 The radius of the circle (2)
- 4.2.2  $TR$  (2)
- 4.3 Determine the equation of the tangent to the circle at  $K$ . Write your answer in the form  $y = mx + c$ . (5)
- 4.4 A horizontal line is drawn as a tangent to the circle  $M$  at the point  $N(c; d)$ , where  $d < 0$ .
- 4.4.1 Write down the coordinates of  $N$ . (2)
- 4.4.2 Determine the equation of the circle centred at  $N$  and passing through  $T$ . Write your answer in the form  $(x - a)^2 + (y - b)^2 = r^2$ . (3)

**[20]**

**QUESTION 5**

- 5.1 In the diagram below,  $P(-7 ; 4)$  is a point in the Cartesian plane.  $R$  is a point on the positive  $x$ -axis such that obtuse  $\hat{POR} = \theta$ .



Calculate, **without using a calculator**, the:

- 5.1.1 Length OP (2)
- 5.1.2 Value of:
- (a)  $\tan \theta$  (1)
- (b)  $\cos(\theta - 180^\circ)$  (2)
- 5.2 Determine the general solution of:  $\sin x \cos x + \sin x = 3 \cos^2 x + 3 \cos x$  (7)
- 5.3 Given the identity:  $\frac{\sin 3x}{1 - \cos 3x} = \frac{1 + \cos 3x}{\sin 3x}$
- 5.3.1 Prove the identity given above. (3)
- 5.3.2 Determine the values of  $x$ , in the interval  $x \in [0^\circ ; 60^\circ]$ , for which the identity will be undefined. (3)

**[18]**



**QUESTION 6**

- 6.1 **Without using a calculator**, simplify the following expression to a single trigonometric term:

$$\frac{\sin 10^\circ}{\cos 440^\circ} + \tan(360^\circ - \theta) \cdot \sin 2\theta \quad (6)$$

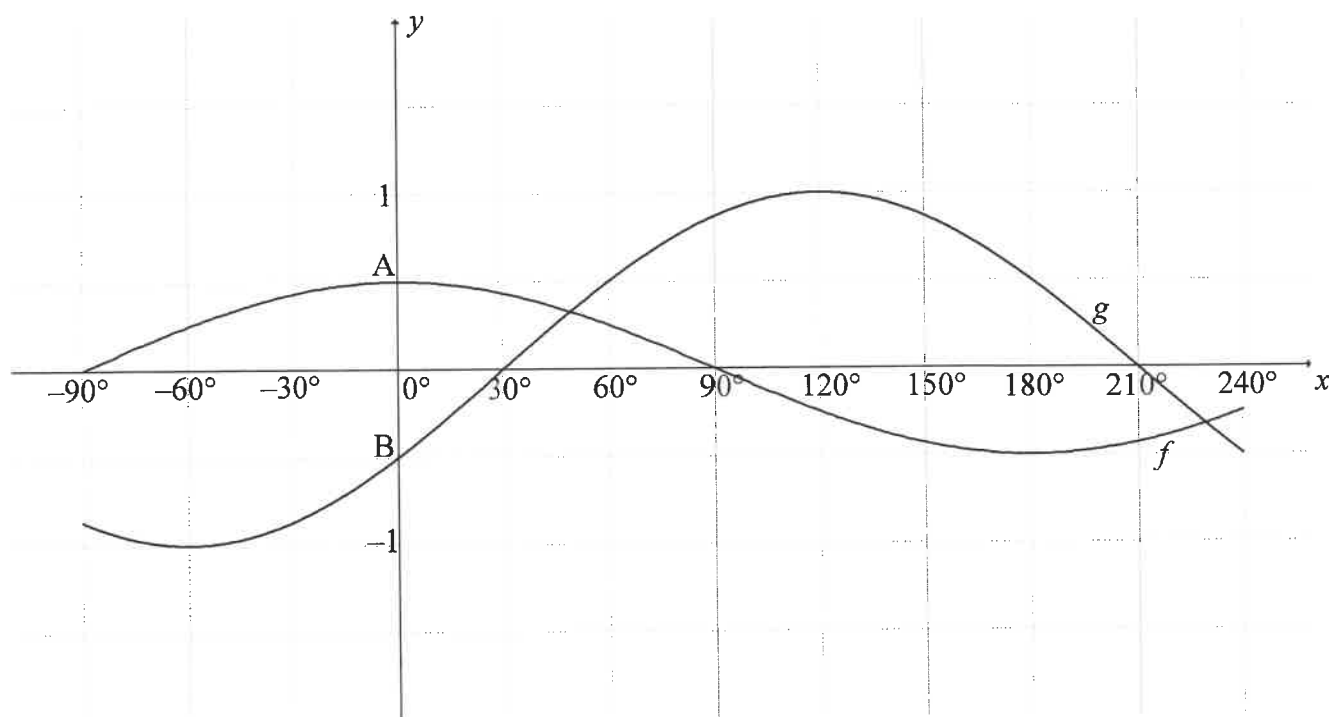
- 6.2 Given:  $\sin(60^\circ + 2x) + \sin(60^\circ - 2x)$

6.2.1 Calculate the value of  $k$  if  $\sin(60^\circ + 2x) + \sin(60^\circ - 2x) = k \cos 2x$ . (3)

6.2.2 If  $\cos x = \sqrt{t}$ , **without using a calculator**, determine the value of  $\tan 60^\circ [\sin(60^\circ + 2x) + \sin(60^\circ - 2x)]$  in terms of  $t$ . (3)  
**[12]**

**QUESTION 7**

In the diagram below, the graphs of  $f(x) = \frac{1}{2}\cos x$  and  $g(x) = \sin(x - 30^\circ)$  are drawn for the interval  $x \in [-90^\circ; 240^\circ]$ . A and B are the y-intercepts of  $f$  and  $g$  respectively.



7.1 Determine the length of AB. (2)

7.2 Write down the range of  $3f(x) + 2$ . (2)

7.3 Read off from the graphs a value of  $x$  for which  $g(x) - f(x) = \frac{\sqrt{3}}{2}$ . (2)

7.4 For which values of  $x$ , in the interval  $x \in [-90^\circ; 240^\circ]$ , will:

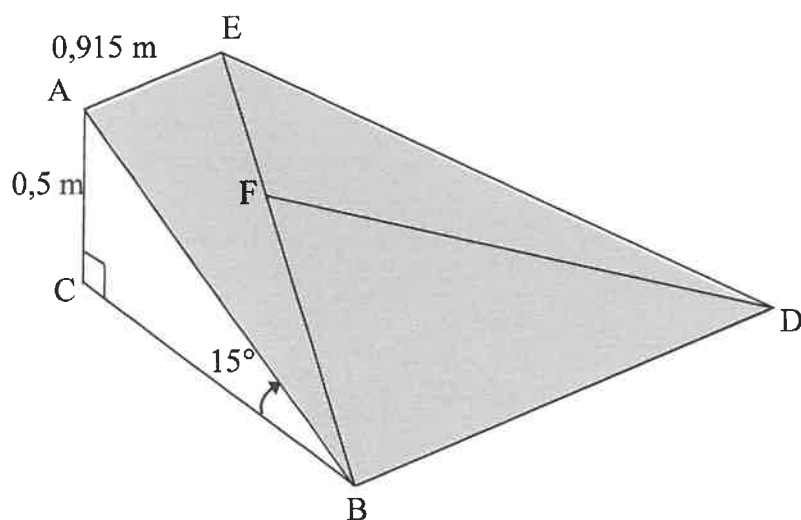
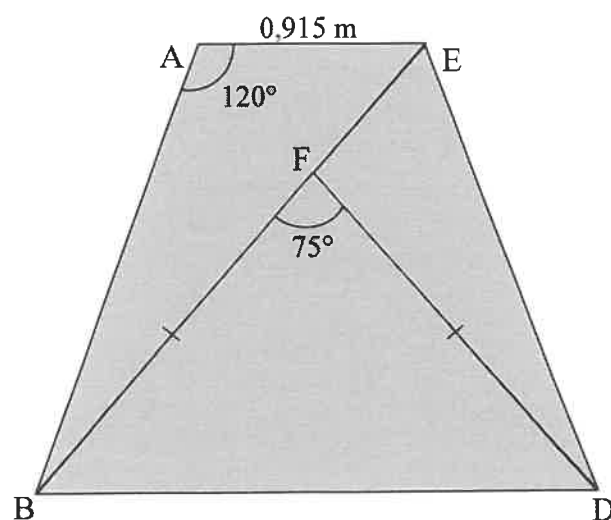
7.4.1  $f(x), g(x) > 0$  (2)

7.4.2  $g'(x - 5^\circ) > 0$  (2)

**[10]**

**QUESTION 8**

FIGURE I shows a ramp leading to the entrance of a building. B, C and D lie on the same horizontal plane. The perpendicular height (AC) of the ramp is 0,5 m and the angle of elevation from B to A is  $15^\circ$ . The entrance of the building (AE) is 0,915 m wide.

**FIGURE I****FIGURE II (top view)**

8.1 Calculate the length of AB. (2)

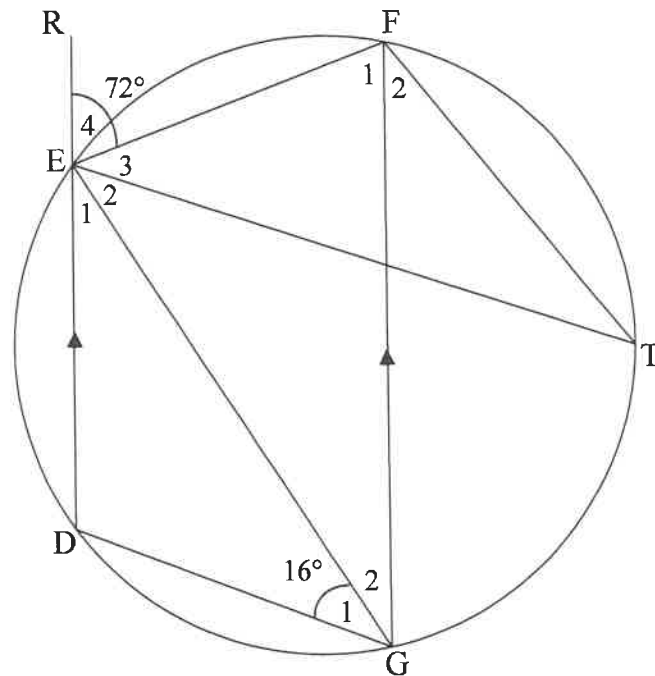
8.2 Figure II shows the top view of the ramp. The area of the top of the ramp is divided into three triangles, as shown in the diagram.

If  $\hat{BAE} = 120^\circ$ , calculate the length of BE. (3)

8.3 Calculate the area of  $\triangle BFD$  if  $\hat{BFD} = 75^\circ$ ,  $BF = FD$  and  $BF = \frac{5}{7}BE$ . (3)  
[8]

**QUESTION 9**

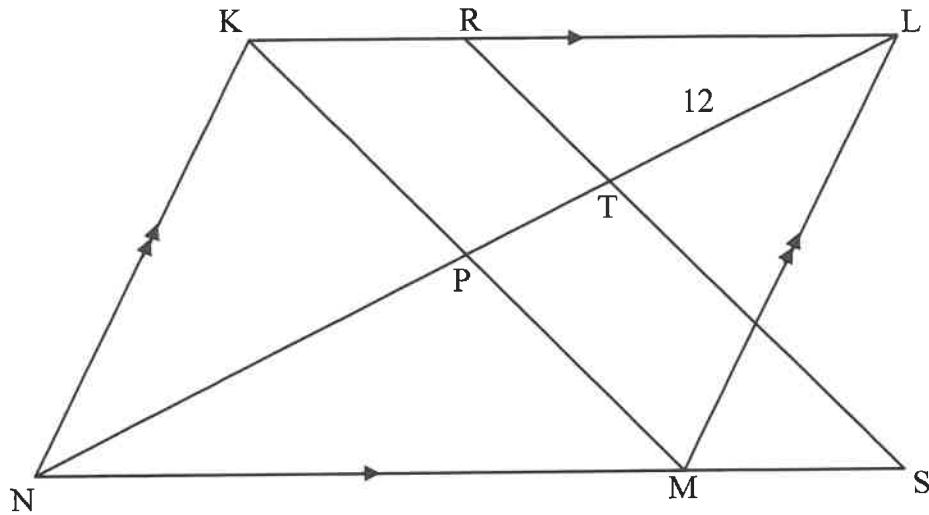
- 9.1 In the diagram, DEFG is a cyclic quadrilateral with  $DE \parallel GF$ . DE is produced to R. T is another point on the circle. EG, FT and ET are drawn.  $\hat{E}_4 = 72^\circ$  and  $\hat{G}_1 = 16^\circ$ .



Determine, with reasons, the size of the following angles:

- |       |             |     |
|-------|-------------|-----|
| 9.1.1 | $\hat{DGF}$ | (2) |
| 9.1.2 | $\hat{T}$   | (2) |
| 9.1.3 | $\hat{GEF}$ | (2) |

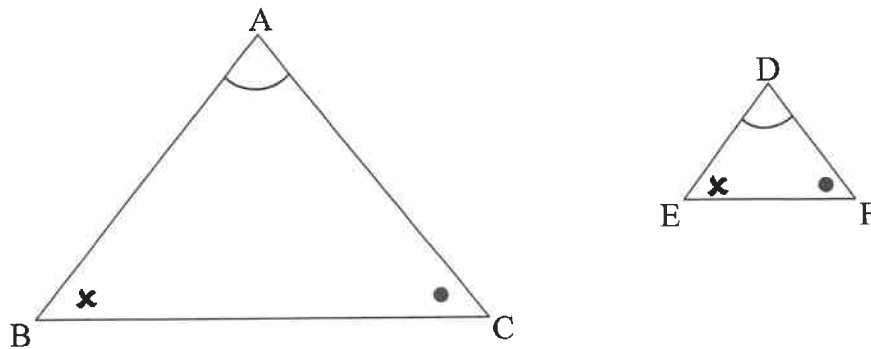
- 9.2 In the diagram, the diagonals of parallelogram KLMN intersect at P. NM is produced to S. R is a point on KL and RS cuts PL at T.  $NM : MS = 4 : 1$ ,  $NL = 32$  units and  $TL = 12$  units.



- 9.2.1 Determine, with reasons, the value of the ratio  $NP : PT$  in simplest form. (4)
- 9.2.2 Prove, with reasons, that  $KM \parallel RS$ . (2)
- 9.2.3 If  $NM = 21$  units, determine, with reasons, the length of  $RL$ . (4)
- [16]**

**QUESTION 10**

10.1 In the diagram,  $\triangle ABC$  and  $\triangle DEF$  are drawn such that  $\hat{A} = \hat{D}$ ,  $\hat{B} = \hat{E}$  and  $\hat{C} = \hat{F}$ .

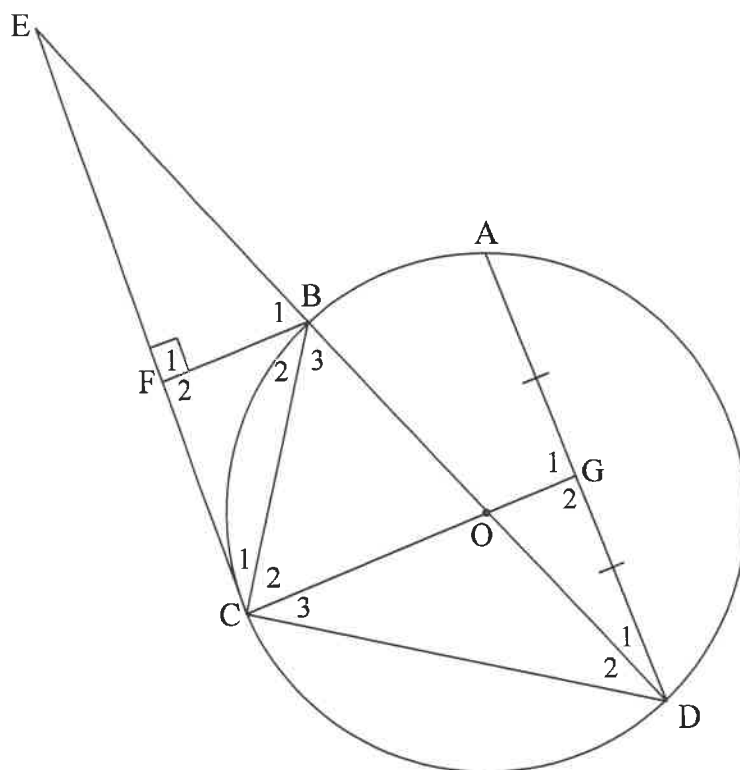


Use the diagram in the ANSWER BOOK to prove the theorem which states that if two triangles are equiangular, then the corresponding sides are in proportion,

i.e.  $\frac{AB}{DE} = \frac{AC}{DF}$ .

(6)

- 10.2 In the diagram, O is the centre of a circle passing through A, B, C and D. EC is a tangent to the circle at C. Diameter DB produced meets tangent EC at E. F is a point on EC such that  $BF \perp EC$ . Radius CO produced bisects AD at G. BC and CD are drawn.



10.2.1 Prove, with reasons, that:

(a)  $FB \parallel CG$  (3)

(b)  $\triangle FCB \parallel \triangle CDB$  (5)

10.2.2 Give a reason why  $\hat{G}_1 = 90^\circ$ . (1)

10.2.3 Prove, with reasons, that  $CD^2 = CG \cdot DB$ . (5)

10.2.4 Hence, prove that  $DB = CG + FB$ . (5)  
[25]

**TOTAL: 150**

## INFORMATION SHEET

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2}ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \cos \alpha \cdot \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cdot \cos \beta - \cos \alpha \cdot \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cdot \cos \beta + \sin \alpha \cdot \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2\sin^2 \alpha \\ 2\cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2 \sin \alpha \cdot \cos \alpha$$

$$\bar{x} = \frac{\sum x}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$